

AI Planning

Introduction. Planning architecture.



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Acknowledgements

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I would like to gratefully acknowledge Dana Nau's contributions and thank him for generously permitting me to use aspects of his presentation material.

Introduction: some Dictionary Definitions of "Plan"



plan *n.*

1. A scheme, program, or method worked out beforehand for the accomplishment of an objective: *a plan of attack.*
 2. A proposed or tentative project or course of action: *had no plans for the evening.*
 3. A systematic arrangement of elements or important parts; a configuration or outline: *a seating plan; the plan of a story.*
 4. A drawing or diagram made to scale showing the structure or arrangement of something.
 5. A program or policy stipulating a service or benefit: *a pension plan.*
- These two are closest to the meaning used in AI

[a representation] of future behavior ... usually a set of actions, with temporal and other constraints on them, for execution by some agent or agents. - Austin Tate

[MIT Encyclopedia of the Cognitive Sciences, 1999]

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A portion of a manufacturing process plan

Introduction. Generating Plans of Action.

- Computer programs to aid human planners
 - Project management (consumer software)
 - Plan storage and retrieval
 - e.g., **variant process planning** in manufacturing
 - Automatic schedule generation
 - various **OR** and AI techniques
- For some problems, we would like generate plans (or pieces of plans) automatically
 - Much more difficult
 - Automated-planning research is starting to pay off



PROCESS PLAN					
OP	A	TIME	Setup	Run time	OW
DEFINITION					
02					Clamp board
001 B	VMC1	0.10	0.43		02 Install 0.09-diameter drill bit
001 C	VMC1	0.10	0.77		02 Rough drill at (1.25, -0.50) to depth
					02 Finish drill at (1.25, -0.50) to depth
					02 Rough drill at (0.00, 4.88) to depth
					03 Finish drill at (0.00, 4.88) to depth
001 T	VMC1	2.20	1.20		01 Total time on VMC1
004 A	VMC1	2.00	0.00		01 Orient board
					02 Clamp board
004 B	VMC1	0.10	0.34		01 Install 0.15-diameter side-milling tool
					02 Rough side-mill pocket at (-0.25, 0.50)
					03 Finish side-mill pocket at (-0.25, 0.50)
					04 Rough side-mill pocket at (-0.25, 3.30)
					05 Finish side-mill pocket at (-0.25, 3.30)
004 C	VMC1	0.10	1.54		01 Install 0.09-diameter end-milling tool
004 T	VMC1	2.50	4.87		01 Total time on VMC1
005 A	EC1	0.00	32.29		01 Pre-clean board (scrub and wash)
					02 Dry board in oven at 85 deg. F
005 B	EC1	30.00	0.48		01 Setup
					02 Screen photoresist from 18000 RPM spin
005 C	EC1	30.00	2.00		01 Setup
					02 Photoimaging of photoresist
005 D	EC1	30.00	20.00		01 Setup
005 T	EC1	90.00	54.77		01 Total time on EC1
006 A	MC1	30.00	4.57		01 Setup
					02 Preheat board for soldering
006 B	MC1	30.00	0.29		01 Setup
					02 Screenprint solder stop on board
006 C	MC1	30.00	7.50		02 Deposit solder paste at (3.35, 1.23)
					31 Deposit solder paste at (3.52, 4.00)
006 T	MC1	90.00	12.36		01 Total time on MC1
007 A	EC1	0.00	35.09		01 Perform postcure reaction of board
007 T	EC1	0.00	64.67		01 Total time on EC1
999 T		319.70	403.37		01 Total time to manufacture

Introduction. Space Exploration.

- Autonomous planning, scheduling, control
 - NASA: JPL and Ames
- Remote Agent Experiment (RAX)
 - Deep Space 1
- Mars Exploration Rover (MER)

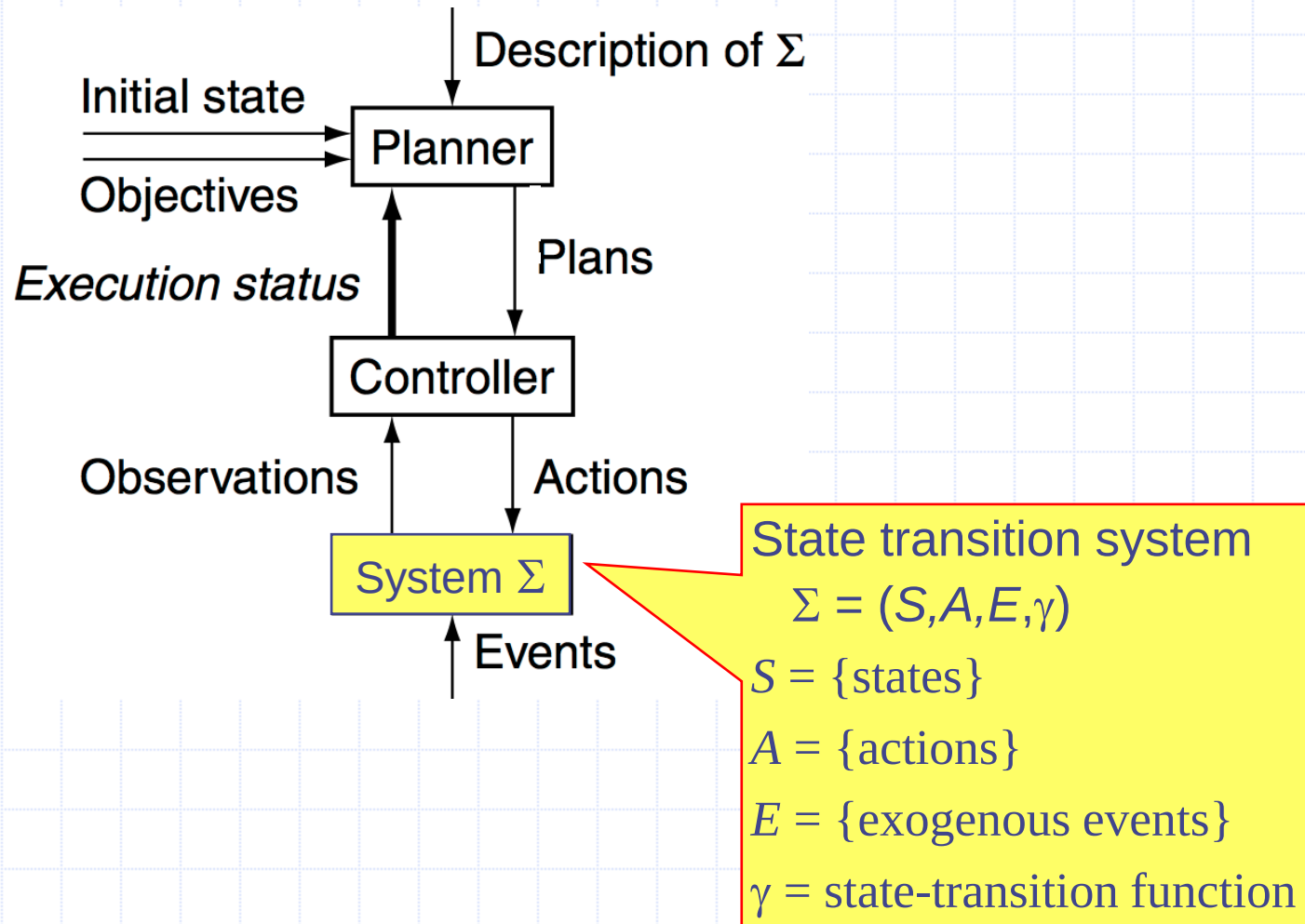


Introduction. Manufacturing.

- Sheet-metal bending machines - Amada Corporation
 - Software to plan the sequence of bends
[Gupta and Bourne, *J. Manufacturing Sci. and Engr.*, 1999]



Introduction: Conceptual Model. Environment.



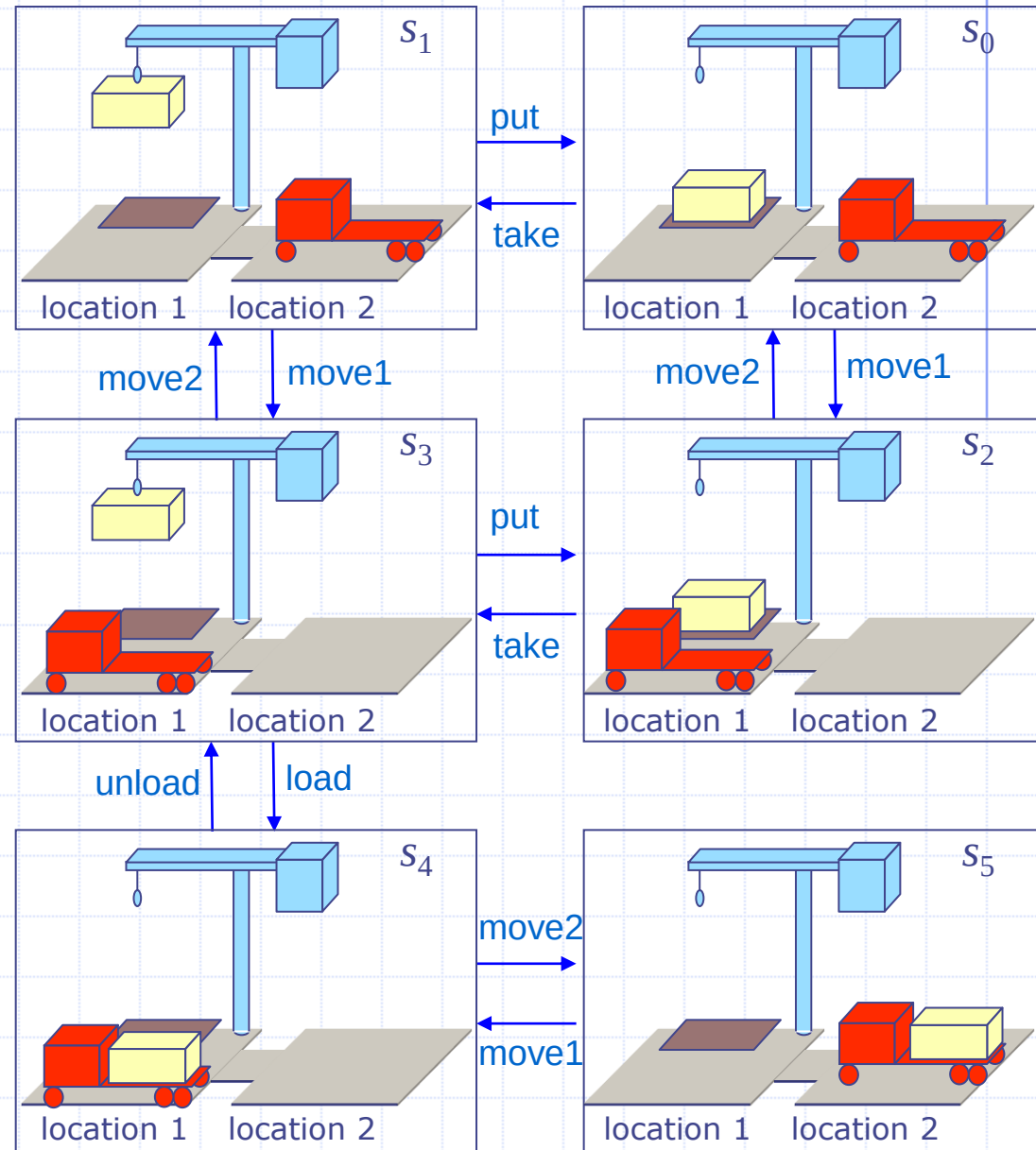
State Transition System

$$\Sigma = (S, A, E, \gamma)$$

- $S = \{\text{states}\}$
- $A = \{\text{actions}\}$
- $E = \{\text{exogenous events}\}$
- State-transition function

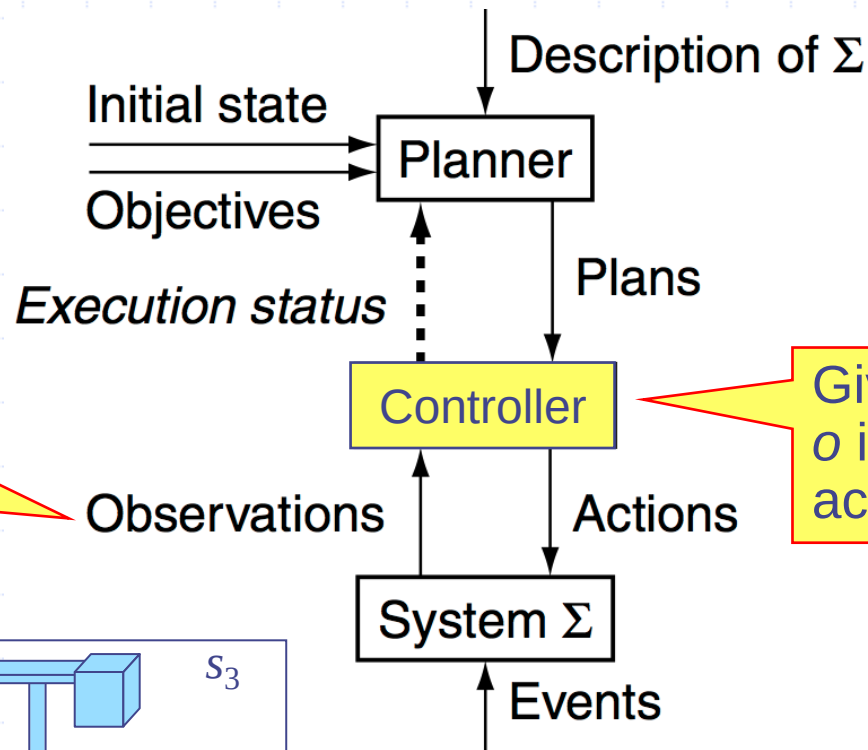
$$\gamma: S \times (A \cup E) \rightarrow S$$

- $S = \{s_0, \dots, s_5\}$
- $A = \{\text{move1, move2, put, take, load, unload}\}$
- $E = \{\}$
- γ : see the arrows



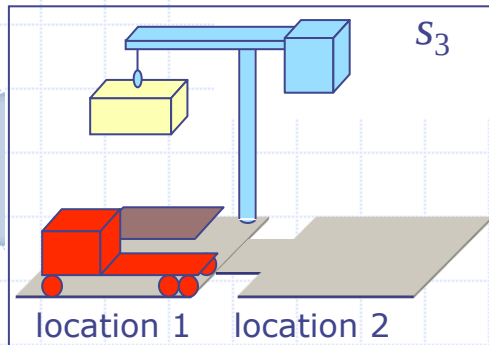
The Dock Worker Robots (DWR) domain

Introduction. Conceptual Model. Controller.



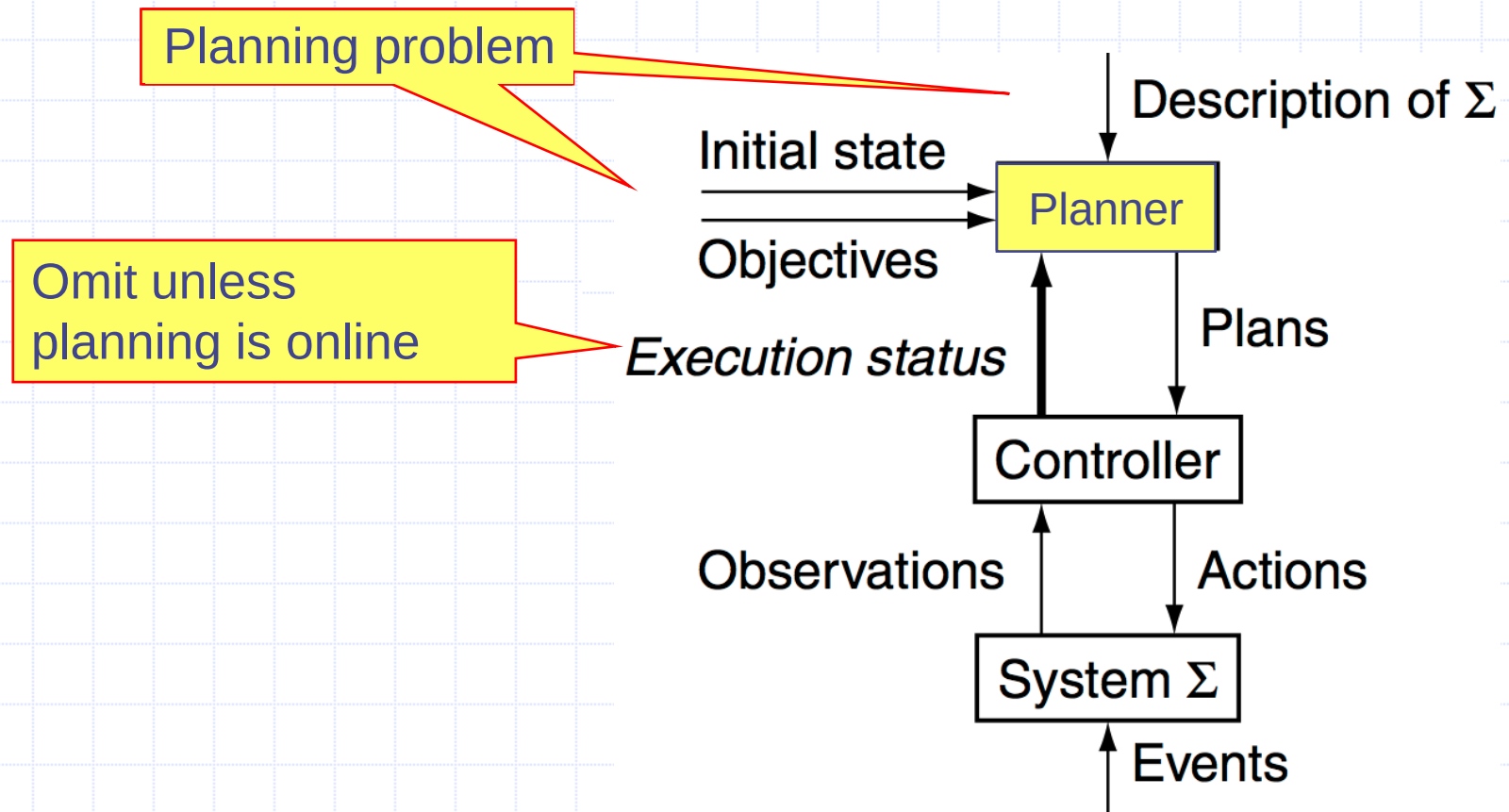
Observation function
 $h: S \rightarrow O$

Given observation
 o in O , produces
action a in A



Introduction.

Conceptual model. Planners input.



Introduction. Planning Problem

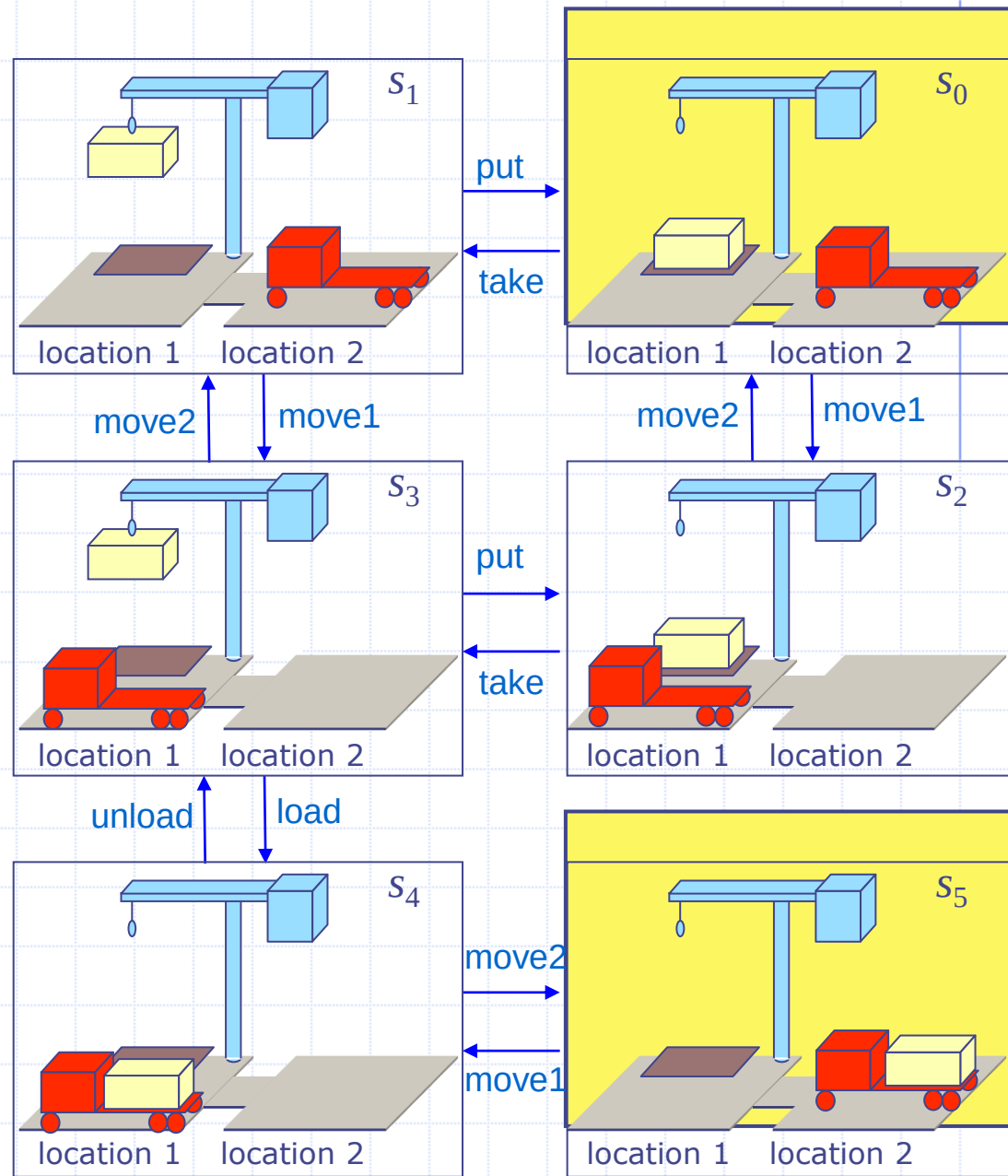
Description of Σ
Initial state or set of states

Initial state = s_0

Objective

Goal state, set of goal states, set of tasks, "trajectory" of states, objective function, ...

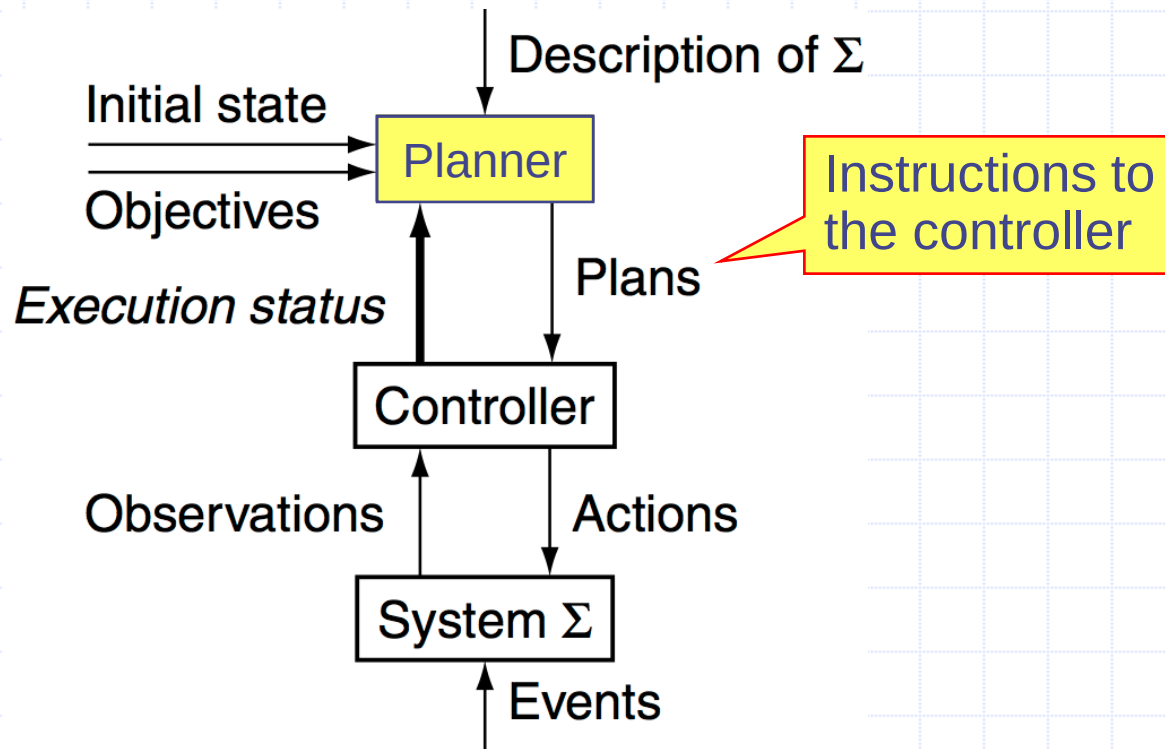
Goal state = s_5



The Dock Worker Robots (DWR) domain

Introduction.

Conceptual Model. Planner's Output.



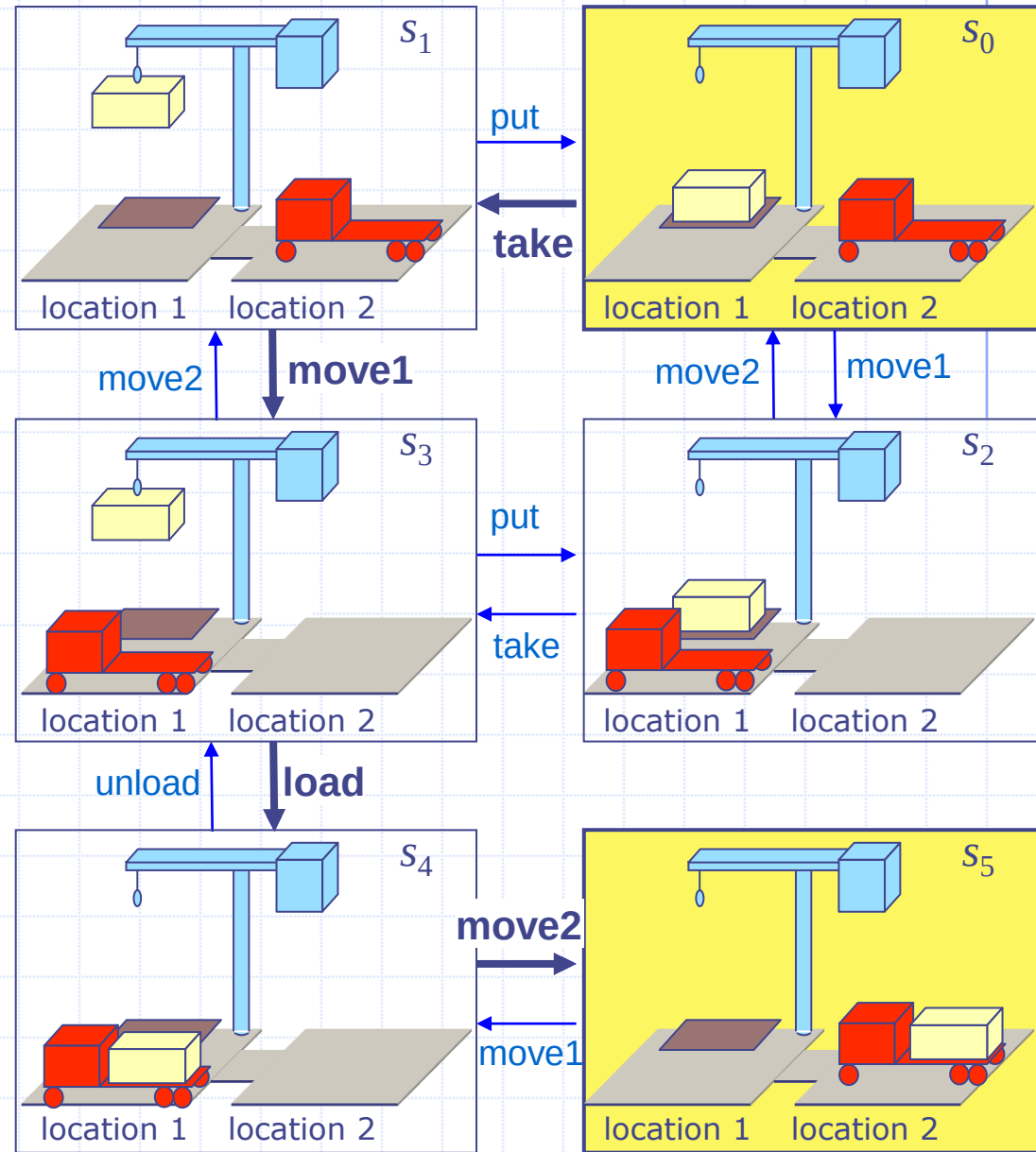
Introduction. Plans

Classical plan: a sequence of actions

$\langle \text{take}, \text{move1}, \text{load}, \text{move2} \rangle$

Policy: partial function from S into A

$\{(s_0, \text{take}),$
 $(s_1, \text{move1}),$
 $(s_3, \text{load}),$
 $(s_4, \text{move2})\}$



The Dock Worker Robots (DWR) domain

Introduction.

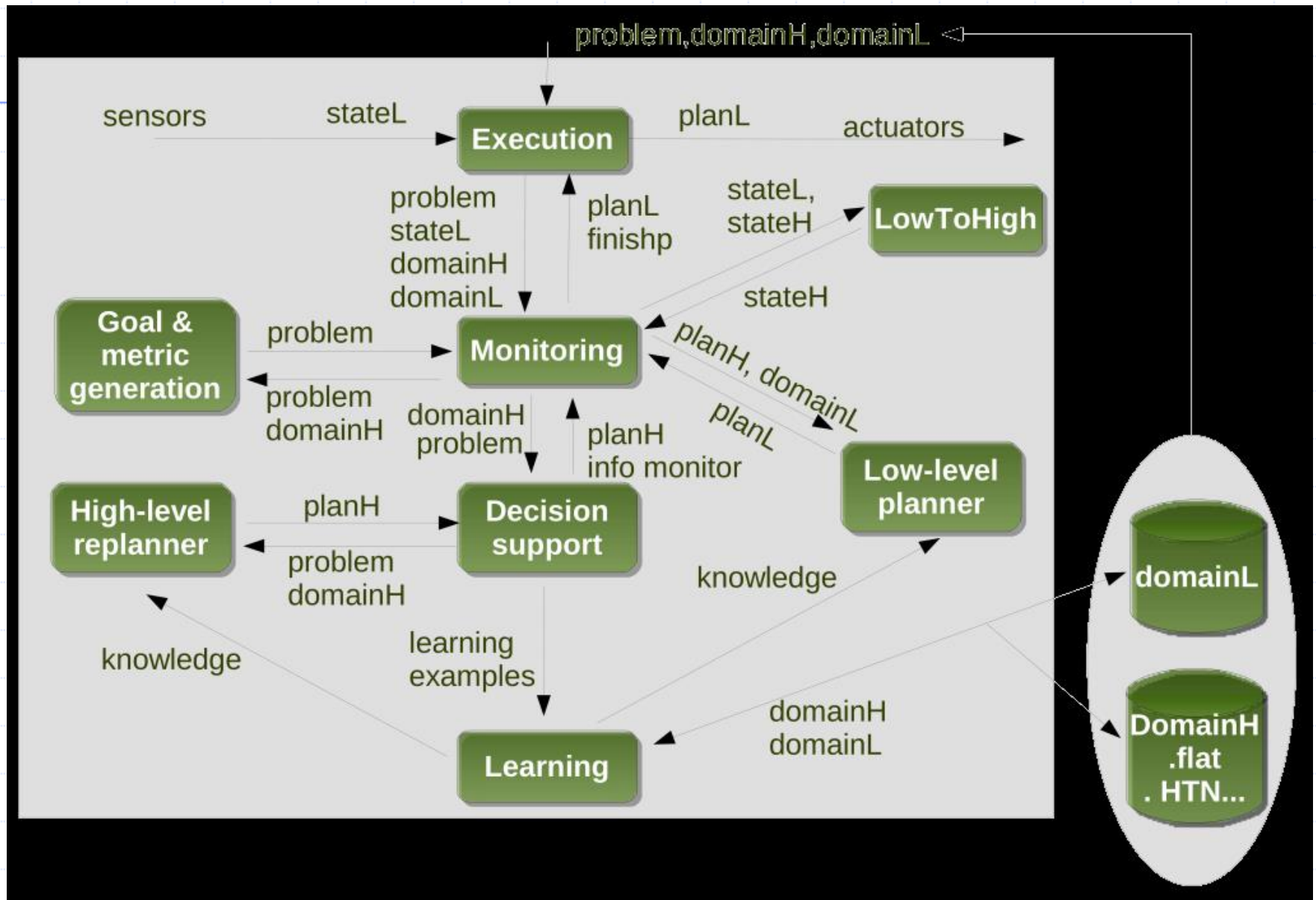
Planning versus Scheduling

- Scheduling
 - Decide when and how to perform a given set of actions
 - Time constraints
 - Resource constraints
 - Objective functions
 - Typically NP-complete
- Planning
 - Decide what actions to use to achieve some set of objectives
 - Can be much worse than NP-complete; worst case is undecidable

Planning architecture: PELEA project

- Integration of the full spectrum of planning-related technologies (sensing, planning, executing, monitoring, replanning and even learning from past experiences).
- PELEA (Planning, Execution and LEarning Architecture) is a general-purpose architecture.
- Control flow, connections among the main PELEA components.

Planning architecture: PELEA project



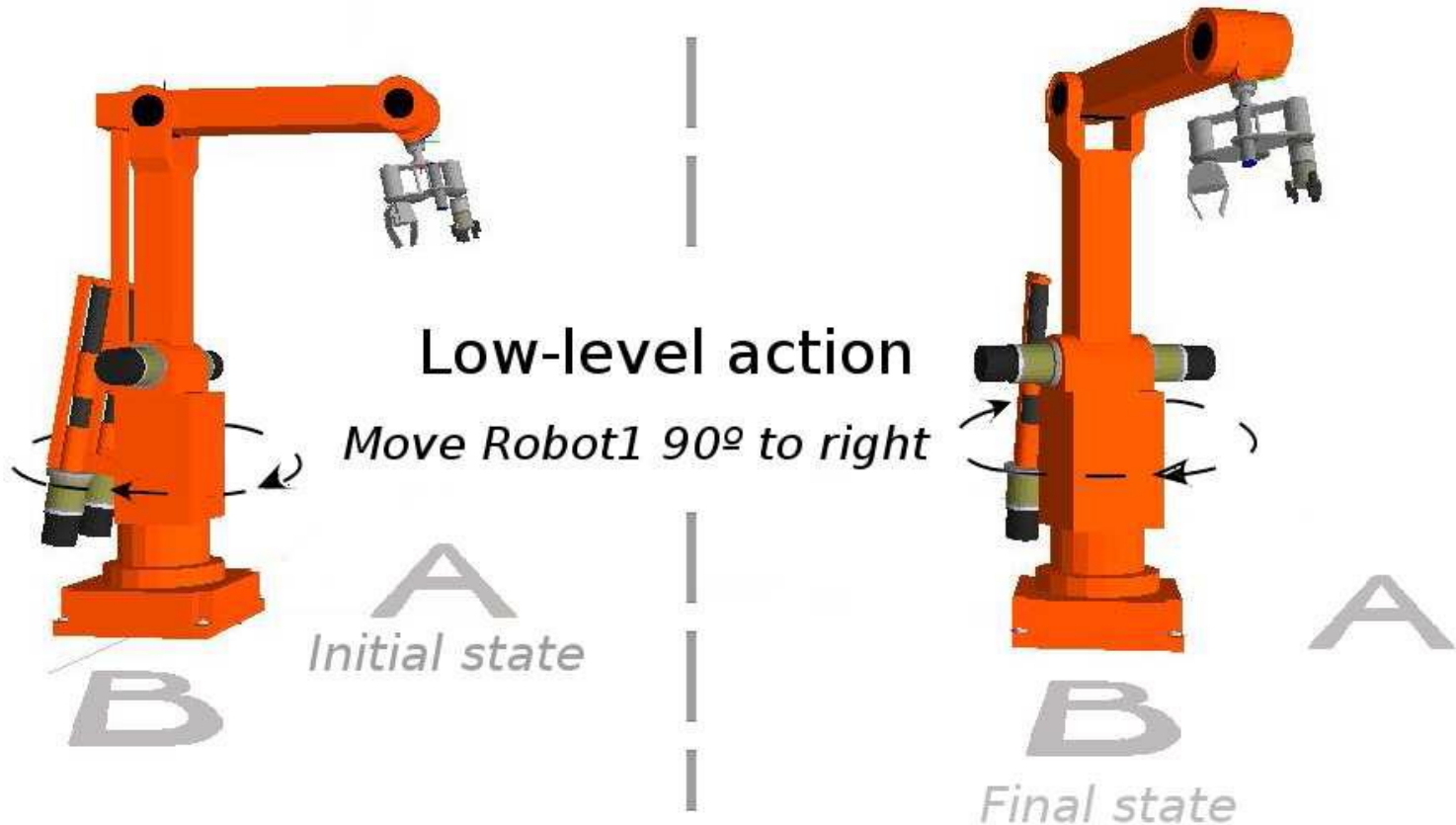
PELEA: the Execution Module (EM)

- EM acts as a wrapper over anything external to PELEA (a software simulator, application, a hardware device like a robot or a user)
- The tasks of the EM are:
 - to initiate PELEA by receiving a particular domain and problem to be solved
 - to observe the current world information by reading the sensors
 - to send the low-level actions to the actuators (for example, move the robot 90 degrees to the right)

PELEA: low-level

High-level action

MOVE ROBOT1 A B



PELEA: the Monitoring Module (MM)

- Check whether the observed state coming from the Execution is a correct state according to the planning process.
- With the info monitor parameter the MM knows which aspects (variables) of the plan need to be monitored during the execution.
- The MM acts as a plan dispatcher sending the Execution Module the actions to execute.

PELEA: the Decision Support (DS) module

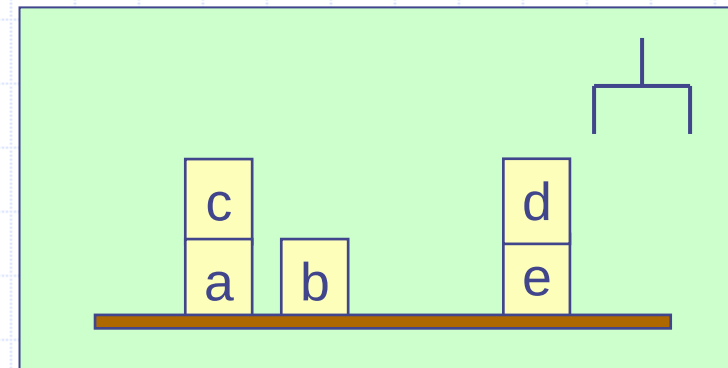
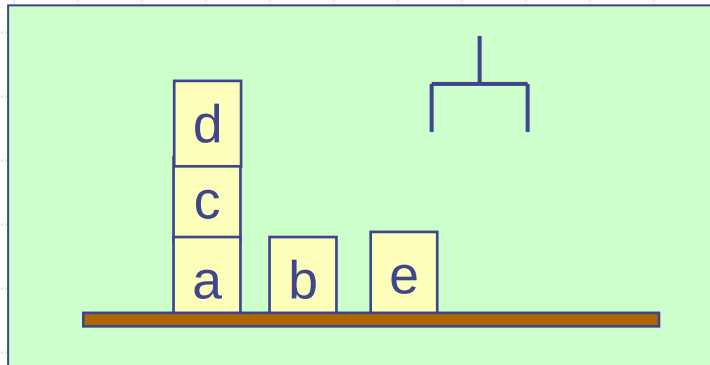
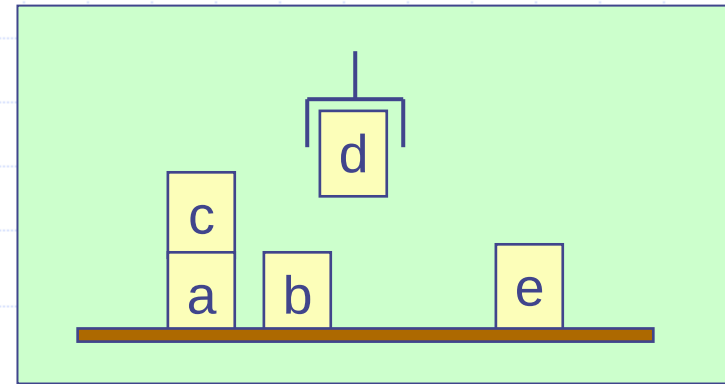
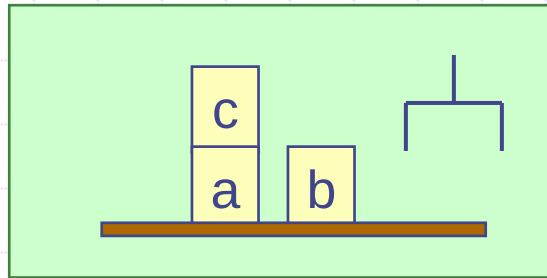
- The DS is responsible for calculating the info monitor parameter.
- When the MM reports a discrepancy between the observed state and the expected planning state, the DS makes the following tasks:
 - to check whether the discrepancy is relevant to the planvalidity or not
 - to make a decision about applying a plan-repair method or replanning from scratch (try to minimizing the number of changes in the original plan)

PELEA: the Planner

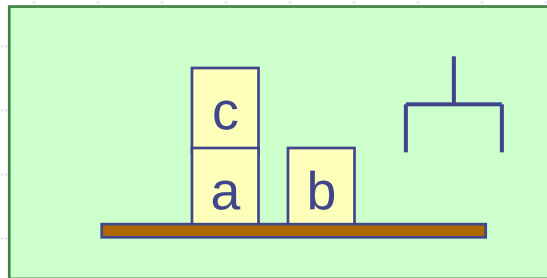
- Planning, repairing and replanning
- We can use any type of planner:
 - State-space planners (plan as a sequence of actions)
 - Plan-space planners (plan as a set of parallel actions)
 - Temporal planners
 - Other type of planners

PELEA: Demo of the PELEA architecture (I)

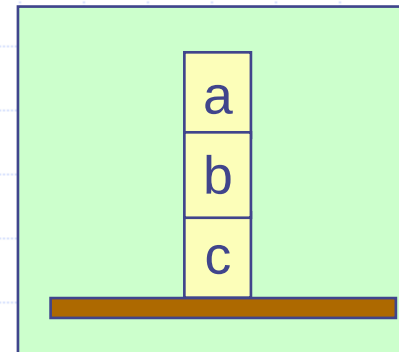
Blocks-world domain



PELEA: Demo of the PELEA architecture (II)



Initial state



Goal state

PELEA: Demo of the PELEA architecture (III)

OPERATORS:

unstack <block> <block>
stack <block><block>
pickup <block>
drop <block>

INSTANTIATED OPERATORS (actions):

- 1. unstack C A**
- 2. drop C**
- 3. pickup B**
- 4. stack B C**
- 5. pickup A**
- 6. stack A B**

plan of actions